

11 The diagram shows a roof support.

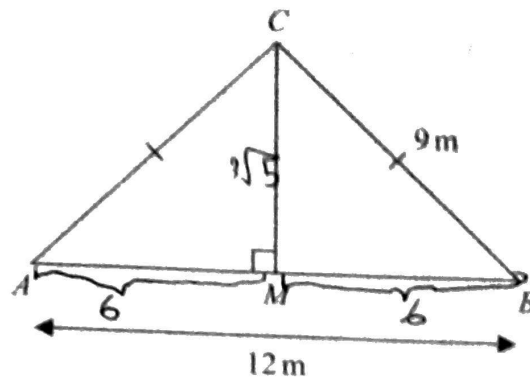


Diagram NOT
accurately drawn

The roof support is made from four lengths of wood, AB, AC, BC and MC

$$AC = BC = 9\text{m} \quad AB = 12\text{m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

M is in between AB

$$AM = MB = \frac{12}{2} = 6$$

Pythagoras Theorem,

$$CB^2 = CM^2 + MB^2$$

$$CM = \sqrt{9^2 - 6^2} = 3\sqrt{5}$$

$$\begin{aligned} \therefore \text{Perimeter of wood} &= 12 + 9 + 3\sqrt{5} + 9 \\ &= 30 + 3\sqrt{5} \text{ m} \end{aligned}$$

$$\therefore \text{total cost} = 21.50 \times 30 + 3\sqrt{5}$$

$$= 789.276$$

$$= \underline{\underline{789.23}}$$

789.23 euros

(Total for Question 11 is 4 marks)

